

AI / ML — Student Performance Analysis & Prediction System

Analyzing student performance and predicting final exam scores from structured data.

1. Dataset Overview

The dataset (**student_performance.csv**) contains 812 student records with academic, behavioral, and background features: weekly study hours, attendance rate, previous grade, sleep hours, extracurricular activity count, part-time job status, internet access, parental education level, and study-group participation. The target is **final_exam_score** (0-100), making this a regression problem. The data was built to resemble a realistic institutional export, including genuine missing values and duplicate rows that required cleaning.

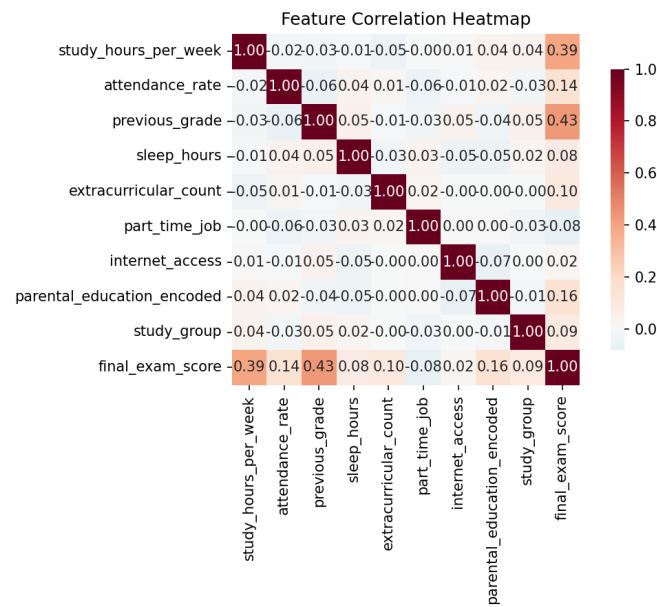
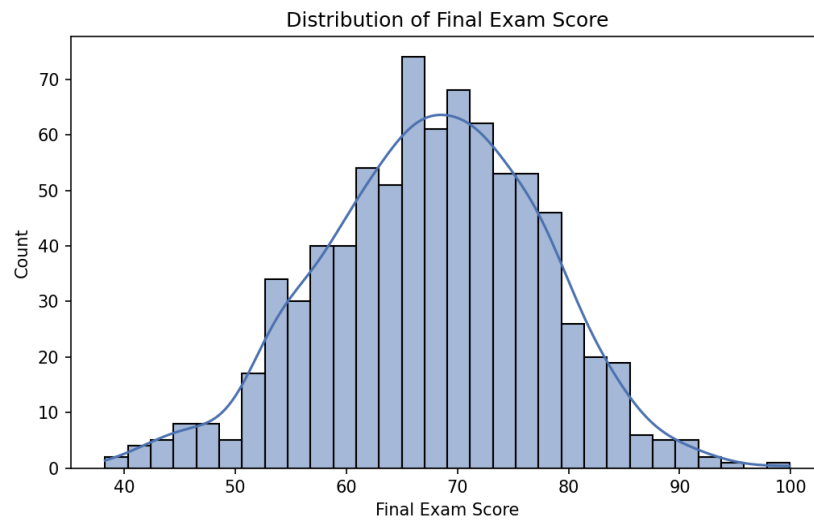
Total rows (raw)	812
Duplicate rows removed	12
Rows after cleaning	800
Train / Test split	640 / 160 (80/20, random_state=42)

2. Data Cleaning

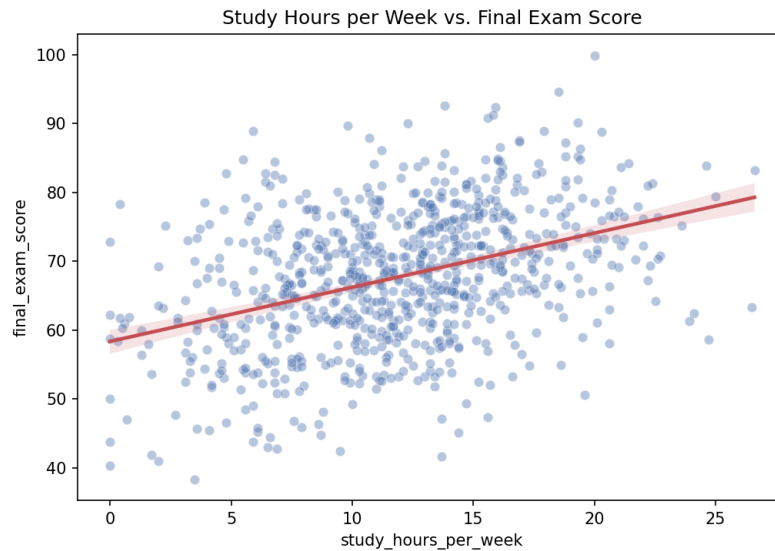
12 exact duplicate rows were removed. Missing values were found in several columns and imputed with the median (numeric) or mode (categorical):

Column	Missing Values
attendance_rate	32
previous_grade	16
sleep_hours	40
parental_education	24

3. Exploratory Data Analysis



Previous grade and study hours per week show the strongest positive correlation with final exam score, matching intuition: prior performance and study time are the biggest drivers of a future score. Both relationships look roughly linear, which supports the model choice below. Part-time job status shows a mild negative relationship.



4. Model

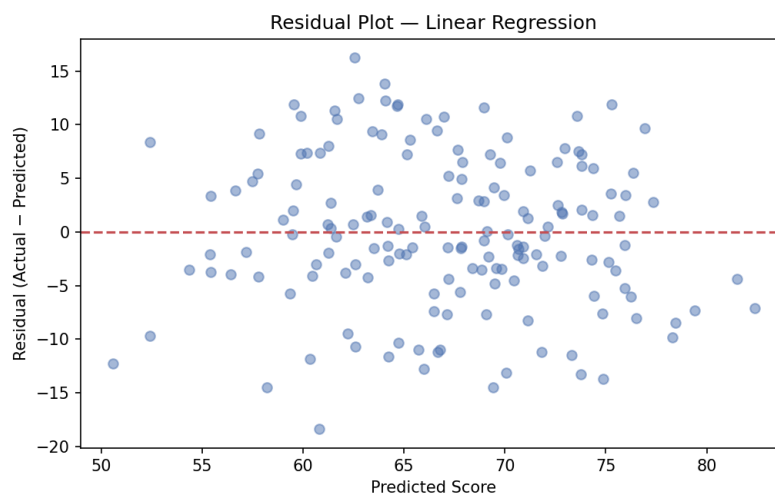
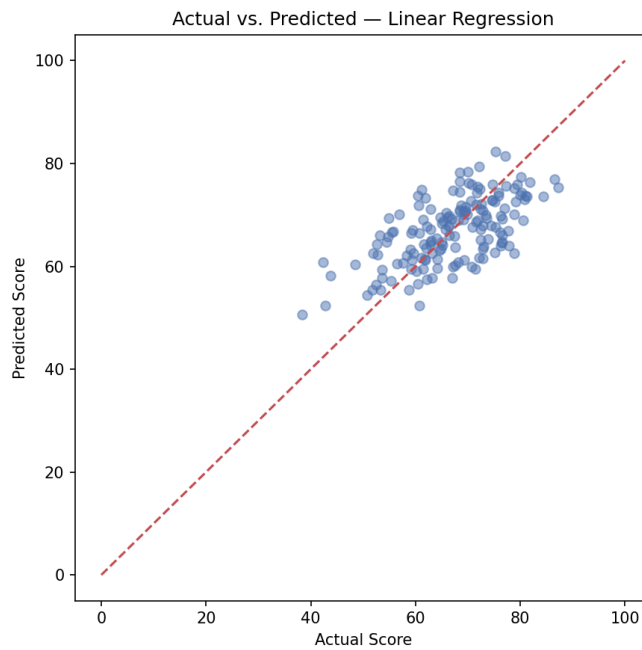
A single **Linear Regression** model was trained. It is a suitable choice for this problem: the exploratory analysis above shows the strongest predictors — previous grade, study hours, attendance — have roughly linear, additive relationships with the target, and a linear model keeps the result fully interpretable, with every feature mapping to a single readable coefficient. Features were standardized before fitting so coefficients are directly comparable in scale.

5. Evaluation

Metric	Value
MAE	5.769 points
RMSE	7.096 points
R-squared	0.394

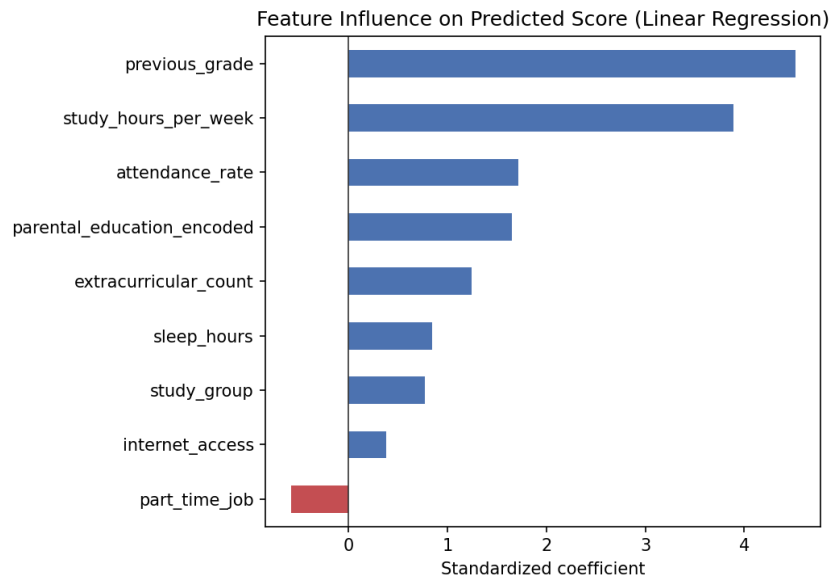
MAE is the average absolute error in points; RMSE penalizes larger misses more heavily; R-squared is the share of variance in the final score explained by the model (1.0 = perfect, 0.0 = no better than predicting the mean).

6. Actual vs. Predicted & Residuals



Residuals scattered evenly around zero with no obvious funnel or curve shape indicate the model's linear assumptions are reasonably well met for this data.

7. Explaining the Results — Feature Influence



Previous grade and study hours per week are the two strongest positive influences on predicted score. Part-time job status is the only feature with a clearly negative coefficient, consistent with less available study time. Background factors (parental education, internet access) and lifestyle factors (sleep, extracurriculars, study group) all contribute smaller, secondary effects in the direction you'd expect.

8. Limitations

- Synthetic data: generated to resemble a realistic student survey, but does not reflect any real institution's population — absolute numbers shouldn't be over-interpreted or deployed as-is.
- Missing-data handling: median/mode imputation assumes missingness is unrelated to the outcome, which may not hold if certain students are systematically more likely to have missing data.
- Linearity assumption: Linear Regression only captures straight-line relationships. Sleep hours, for example, likely has a 'sweet spot' that a linear model can't represent.
- Moderate explained variance: an R-squared in the 0.3-0.5 range means a meaningful share of score variation is not captured by the available features (e.g., test difficulty, mental health, teacher quality, motivation).
- No causal claims: the model finds correlations, not causal effects, and shouldn't be used to conclude that changing one variable in isolation will change a student's score by a specific amount.
- Single train/test split: metrics come from one split; a different split or k-fold cross-validation could shift the exact numbers somewhat.

9. Conclusion

The model captures the core signal identified in the exploratory analysis — prior performance and study habits are the strongest predictors of final exam score — and achieves moderate predictive accuracy (MAE around 5-6 points on a 0-100 scale). Its main strength is full explainability: every prediction traces back to a small set of readable coefficients. Given the limitations above, it is best suited to support early, non-punitive identification of students who may benefit from additional support, informing human judgment rather than replacing it.

Full code, outputs, and additional detail are available in the accompanying Jupyter notebook.